Savannah Drake

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Education

Emory University Atlanta, GA

B.S in Data and Decision Sciences, Informatics Track *May 2027*

Relevant Coursework: Probability & Statistics, Artificial Intelligence, Linear Algebra, Big Data Visualization, Ethics of AI, Natural Language Processing, Data Structures & Algorithms, Empirical Research Methods & Causal Analysis

Relevant Data Science Experience

Team Lead & Data Analyst Jan 2026 – May 2026

American Airlines(via Emory Data Science Club) *Atlanta, GA*

- Queried 18,000 flight records using Python and SQL to identify patterns driving delays and missed connections
- Established and evaluated a predictive ML model (AUC 0.64) to identify high delay risk across operations
- Designed an interactive Streamlit dashboard that converted technical predictive model outputs and large-scale operational data into user-friendly visualizations and actionable insights

Team Lead & Data Analyst Dec 2025 – May 2026

Invest Atlanta(via Emory AI Data Lab) *Atlanta, GA*

- Analyzed 20k+ rows of geospatial and socioeconomic data to identify patterns of disinvestment and disparities in resource access across Atlanta neighborhoods
- Engineered a composite "need score" using quantile binning to identify high-priority communities and support data-driven investment decisions.
- Created an interactive geospatial dashboard with layered mapping visualizations to identify high-investment-need areas and support Atlanta's Fresh Food Access initiative through 2030; project received recognition for best analytical approach at symposium presentation

Student Technical Support Technician Aug 2025 – May 2026

Emory University Office of Information Technology *Atlanta, GA*

- Applied root cause analysis and technical troubleshooting across 40+ device incidents to identify recurring system failures and improve operational reliability.
- Identified patterns in recurring technical failures and implemented targeted solutions to improve overall system performance and reliability by 38%.
- Translated complex technical issues into clear, actionable guidance for users, improving resolution efficiency and overall user experience.

Projects

Spotify Audio Clustering & Music Insights | *Python, SciKit-learn, K-Means, Streamlit, Data Visualization*May 2026

- Developed a Spotify audio analysis and clustering application by analyzing 10,000+ songs by audio characteristics including energy, danceability, tempo, and valence
- Implemented unsupervised machine learning workflows using K-Means clustering, feature scaling, and dimensional analysis to improve similarity grouping accuracy and reveal interpretable listening patterns across datasets
- Performed end-to-end data preprocessing and exploratory data analysis on large music datasets, cleaning missing values and engineering audio-feature pipelines that improved clustering consistency and visualization quality.
- Designed and deployed an interactive analytics dashboard enabling real-time cluster visualization, dataset uploads, and user-driven exploration of music trends and recommendation behavior.

Technical Skills

Programming & Data: SQL(joins, aggregations), Python(pandas, NumPy, Scikit-learn, geopandas, matplotlib), R
Data Analysis: cleaning, exploratory data analysis(EDA), statistical analysis, feature engineering, machine learning
Visualization Tools: Streamlit, Excel, Power BI(dashboard), Tableau
Data & Workflow Tools: Git/Github, APIs, Data Pipeline Development